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Faculty of Technology and Engineering
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IT475: Data Analytics & Visualization [ELECTIVE-III]
Practical List [2024]

1.	Introduction to Python - Understanding data structures: List, Dictionary, Tuple, Set etc. - Looping over arrays. - Slicing and Updating elements - Functions and classes - Write a Python function that takes a list of customer ages and returns a new list containing only the ages of customers who are 21 years or older. Use a loop to iterate through the list. Test input: [16, 22, 18, 24, 30, 19, 21] Test Output: [22, 24, 30, 21] - Create a dictionary that maps the names of employees to their salaries. Write a function that takes this dictionary and returns the name of the highest-paid employee. Test Input: {'Alice': 70000, 'Bob': 50000, 'Charlie': 85000} Test Output: 'Charlie' - Given a tuple of daily temperatures for a week, write a function that returns a new tuple containing only the temperatures from the second to the second-to-last day. Test Input: (68, 70, 75, 80, 79, 73, 68) Test Output: (70, 75, 80, 79, 73) - Define a class Student that has attributes name and courses (a set of course names). Implement a method add_course to add a course to the student's set of courses and a method get_courses to return the set of courses. Test: student.get_courses() Output: {'Math', 'Science'} - Write a Python program that takes a list of customer names and a corresponding list of their order amounts, then creates a dictionary where the customer names are keys and their order amounts are values. Write a function to print each customer's name and order amount. Test: print_orders(orders) Test Output: Alice: \$250.5 , Bob: \$300.75, Charlie: \$150.0 Help https://learnpython.org/ https://www.kaggle.com/learn/intro-to-programming
2.	Numpy - Create array and perform indexing. - Slicing and Updating elements, - Shape manipulations. - Looping over arrays. - Reading files in numpy - Write a Python function that creates a 1D numpy array containing the numbers from 0 to 9, and then returns the element at index 5. - Create a 2D numpy array with shape (3, 3) containing the numbers from 1 to 9. Write a function to return a slice of the array containing the first two rows and the first two columns. Then, update the element at position (1, 1) to 100. - Write a Python function that takes a 2D numpy array and prints each element along with

	its position in the array. 4) Write a Python function that reads a CSV file containing numerical data into a numpy array and returns the array. Assume the file is named data.csv. Pandas • Creating data frame. • Reading files. • Slicing manipulations. • Exporting data to files. • Columns and row manipulations with loops. • Use pandas for masking data and reading if in Boolean format. 1) Write a Python function that creates a pandas DataFrame with three columns: 'Name', 'Age', and 'City'. The DataFrame should contain the following data: Name: ['Alice', 'Bob', 'Charlie'] Age: [25, 30, 35] City: ['New York', 'Los Angeles', 'Chicago'] 2) Write a Python function that reads a CSV file named data.csv into a pandas DataFrame. 3) Write a Python function that takes a pandas DataFrame and returns a new DataFrame containing only the first two rows and the columns 'Name' and 'Age'. 4) Write a Python function that takes a pandas DataFrame and exports it to a CSV file named output.csv. 5) Write a Python function that takes a pandas DataFrame and prints each row's 'Name' and 'City' using a loop. 6) Write a Python function that reads a CSV file named data.csv into a pandas DataFrame, and returns a new DataFrame containing only the rows where the 'Age' column is greater than 28. Help ❖ https://www.hackerearth.com/practice/machine-learning/data-manipulation-visualisation-r-python/tutorial-data-manipulation-numpy-pandas-python/tutorial/ ❖ https://www.dataquest.io/m/289-introduction-to-numpy/
3.	Case study based on Numpy and pandas Select the Dataset of your choice and respond to the following questions. Content of case study should include the following points. - Introduction and Dataset Description. - Data Loading and Initial Exploration - Data Cleaning and Preprocessing - Data analysis and operations. - Generating Insights and Reporting Help ❖ https://www.kaggle.com/learn/data-cleaning ❖ https://www.kaggle.com/search?q=data+analytics+in%3Adatasets
4.	Charts with Matplotlib Select the Dataset of your choice and create different types of charts as mentioned below. You can explore dataset from https://data.gov.in/ - Basic Charts (Line Chart, Scatter plots, Bar chart, Pie Chart, Histogram, Boxplot) - Advanced Charts (Violin Plot, Error bars, Contour Plot, Surface Plot) - Matrix Plots (Heat map) Sample dataset: https://data.gov.in/catalog/district-wise-students-institutions-punjab Expected tasks: • Write a Python function to create a line chart showing the number of boys in other institutions across different years for each district.

	• Write a Python function to create a scatter plot showing the number of boys in 2017 versus 2016 for each district. • Write a Python function to create a bar chart showing the number of boys in 2017 for each district. • Write a Python function to create a pie chart showing the proportion of the total number of boys in 2017 for each district. • Write a Python function to create a histogram of the number of boys in 2017. • A boxplot showing the distribution of the number of boys across the years 2014 to 2017. • Write a Python function to create a violin plot of the number of boys in other institutions from 2014 to 2017. • Write a Python function to create a bar chart with error bars showing the average number of boys in other institutions from 2014 to 2017, with standard deviation as error bars. • Write a Python function to create a contour plot for the number of boys in 2014 and 2017. • Write a Python function to create a 3D surface plot for the number of boys in 2014, 2016, and 2017. • Write a Python function to create a heatmap showing the number of boys in other institutions for each district from 2014 to 2017. 1) Discuss the effectiveness of different chart types in conveying specific types of data (e.g., trends, distributions). Provide examples where each chart type would be most suitable. 2) Compare the use cases of basic charts versus advanced charts in data analysis and visualization. When would you choose to use a violin plot over a boxplot? 3) Explain the significance of using matrix plots like heat maps in exploratory data analysis. How can heat maps help identify patterns or correlations in large datasets? Help ❖ https://matplotlib.org/stable/gallery/showcase/anatomy.html ❖ https://matplotlib.org/stable/gallery/index.html
5.	Charts with Seaborn Select the Dataset of your choice and create different types of charts as mentioned below. You can explore dataset from https://data.gov.in/ and perform the tasks as mentioned in practical 4. i. Relational Plots (Scatter plot, Line plot) ii. Categorical Plots (bar plot, boxplot, swarm plot) iii. Time Series Plots(line plot) iv. Statistical Estimation Plots (point plot, bar plot, cat plot) 1) Discuss the importance of scatter plots in identifying correlations between two numerical variables. Provide an example where this visualization revealed an unexpected insight. 2) How do line plots help in understanding trends over time? Discuss the limitations of using line plots for non-time series data. 3) Compare and contrast bar plots, box plots, and swarm plots in terms of their effectiveness in visualizing categorical data. When would you prefer one over the others? Provide examples to support your discussion. 4) How can combining different categorical plots provide a more comprehensive view of the data? Provide an example where using multiple plot types revealed deeper insights. 5) Explain the significance of line plots in time series analysis. How do they help in forecasting and identifying seasonal patterns? Provide an example of a business application where line plots are crucial.

	Help ❖ https://www.kaggle.com/learn/data-visualization
6.	Perform Exploratory Data Analysis using Matplotlib and Seaborn library of dataset. Select the Dataset of your choice and respond to the following questions. 1) What are the dimensions of the dataset? How many rows and columns does it contain? 2) What types of variables are present in the dataset (e.g., numerical, categorical)? Provide a summary of the data types. 3) What are the summary statistics (mean, median, standard deviation) of the numerical columns? 4) What are the minimum, maximum, and quartile values of the key numerical variables? 5) Are there any missing values in the dataset? If so, which columns have missing values and what percentage of the data do they represent? 6) How would you handle the missing values in the dataset? Discuss different strategies and their potential impact on the analysis. 7) Discuss the rationale behind choosing specific types of visualizations (e.g., histograms, scatter plots, heatmaps) for your EDA. How do these visualizations help in understanding the data better? 8) How can combining different visualizations provide a more comprehensive understanding of the dataset? Provide an example where multiple visualizations revealed deeper insights. 9) How do you interpret the correlation matrix and heat map? Discuss any surprising correlations and their potential implications. 10) What steps would you take if you identify high multicollinearity among the variables? How can it affect your analysis? 11) Summarize the key findings from your EDA. What are the main insights you have gained about the dataset? ❖ https://www.kaggle.com/code/ekami66/detailed-exploratory-data-analysis-with-python
7.	Load dataset(s) (e.g., Excel, CSV, database) and perform the operations in Power BI. ❖ Data Modeling ❖ Navigation ❖ Filtering and slicing ❖ Data Exploration and Analysis 1) Discuss the importance of data modeling in Power BI. How does a well-designed data model impact the performance and accuracy of your reports? 2) What challenges might you face when creating relationships between tables, and how can you address them? 3) How does the navigation in Power BI enhance the user experience? Discuss the advantages of having different views (Report, Data, Model) for data analysis and reporting. 4) What tips can you share for efficiently navigating large datasets and complex reports in Power BI? 5) How do filters and slicers improve the interactivity of Power BI reports? Discuss the differences between visual-level, page-level, and report-level filters and their use cases. 6) How do visualizations aid in data exploration and analysis in Power BI? Discuss the process of choosing the right visualization for different types of data. 7) What are the benefits of using advanced AI visuals in Power BI? Provide examples of how these visuals can provide deeper insights compared to traditional charts. Help https://www.tutorialspoint.com/power_bi/index.html

8.	Designing Dashboards in Power BI and storytelling via dashboards. <table border="1" data-bbox="268 230 1401 526"> <tr> <td data-bbox="268 230 818 526"> ❖ Define Objectives ❖ Data Preparation ❖ Create Individual Visualizations. ❖ Design the Dashboard Layout ❖ Implement the Dashboard in Power BI ❖ Test and Refine ❖ Deploy and Share </td><td data-bbox="818 230 1401 526"> ❖ Identify the Story ❖ Structure the Narrative ❖ Choose Appropriate Visuals ❖ Focus on User Engagement ❖ Provide Context and Interpretation ❖ Review and Iterate </td></tr> </table> 1) What are the key design principles you follow when creating a Power BI dashboard? How do these principles help in delivering a clear and impactful message? 2) Discuss the balance between aesthetics and functionality in dashboard design. How do you ensure your dashboard is both visually appealing and highly functional? 3) What are the key design principles you follow when creating a Power BI dashboard? How do these principles help in delivering a clear and impactful message? 4) Discuss the balance between aesthetics and functionality in dashboard design. How do you ensure your dashboard is both visually appealing and highly functional? 5) How do you tailor a Power BI dashboard to different user groups (e.g., executives, analysts, operational staff)? What considerations vary between these audiences? 6) Provide examples of how user feedback has led to significant changes or improvements in your dashboard design. What insights were most valuable? Help https://www.youtube.com/watch?v=UXhGRVTndQA	❖ Define Objectives ❖ Data Preparation ❖ Create Individual Visualizations. ❖ Design the Dashboard Layout ❖ Implement the Dashboard in Power BI ❖ Test and Refine ❖ Deploy and Share	❖ Identify the Story ❖ Structure the Narrative ❖ Choose Appropriate Visuals ❖ Focus on User Engagement ❖ Provide Context and Interpretation ❖ Review and Iterate
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9.	Perform Data Analytics using Power BI using a dataset. Generate a report that contains various visualizations. ❖ Import Data ❖ Clean and Transform Data ❖ Create Data Model ❖ Explore Data ❖ Design Visualizations ❖ Create the Dashboard ❖ Generate the Report ❖ Publish and Share 1) Discuss the importance of data modeling in Power BI. How does a well-designed data model impact the performance and accuracy of your reports? 2) What challenges did you face during data preparation, and how did you overcome them? 3) How did you decide which visualizations to include in your report? Discuss the factors you considered. 4) Provide examples of visualizations that were particularly effective in conveying insights and why. 5) How did interactive features (filters, slicers, drill-throughs) improve the usability of your Power BI report? 6) Discuss any feedback received from users about the interactivity of your report. How did you address their suggestions? 7) What are the most actionable insights derived from your analysis?		
10.	Load dataset(s) (e.g., Excel, CSV, database) and Perform the operations in tableau. ❖ Connect to a Data Source. ❖ Integrate Multiple Data Sources. ❖ Handle Data Extracts. ❖ Data Preparation and Transformation.		

	❖ Create Hierarchies and Groups. ❖ Apply filters and slicers. 1) Discuss the challenges you faced when connecting to and integrating multiple data sources in Tableau. How did you overcome these challenges? 2) What were the most significant data preparation and transformation tasks you performed in Tableau? How did these tasks improve the quality of your data analysis? 3) How do calculated fields and hierarchies enhance the analytical capabilities of Tableau? Provide examples from your experience. 4) How do filters and slicers improve the interactivity and usability of Tableau dashboards? Discuss specific examples where filtering and slicing were crucial. Help https://www.youtube.com/watch?v=3GT_TLIeCbE https://www.geeksforgeeks.org/tableau-tutorial/
11.	Perform Data Analytics using Tableau using dataset. Generate dashboard and report that contains various visualizations. ❖ Prepare your dataset ❖ Connect to Your Data in Tableau. ❖ Create Visualizations. ❖ Deep Dive Analysis ❖ Dashboard Creation ❖ Generate Report 1. How do you design an effective dashboard in Tableau? What are the key elements to consider for layout and storytelling? 2. How do you add and customize interactivity in your Tableau dashboard? Discuss the use of filters, parameters, and dashboard actions. 3. How do you publish and share your Tableau dashboard? What are the options for configuring access and permissions? 4. What are the overall trends over time? 5. How are metrics distributed across different categories or regions? 6. What is the composition of different categories? 7. What are the top or bottom performers in a specific metric? 8. How do different segments compare in terms of a specific metric? 9. What are the patterns in the data over different periods? 10. What is the geographic distribution of a metric? 11. Provide interpretations and possible explanations for observed trends and patterns. https://www.tableau.com/learn/articles/free-public-data-sets